

* PAPER 1 · PHYSICAL CONSTANTS

Quantity	Symbol	Value	Quantity	Symbol	Value
Acceleration due to gravity	g	$9,8 \text{ m}\cdot\text{s}^{-2}$	Speed of light in a vacuum	c	$3,0 \times 10^8 \text{ m}\cdot\text{s}^{-1}$
Universal gravitational constant	G	$6,67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$	Planck's constant	h	$6,63 \times 10^{-34} \text{ J}\cdot\text{s}$
Radius of the Earth	R_E	$6,38 \times 10^6 \text{ m}$	Coulomb's constant	k	$9,0 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$
Mass of the Earth	M_E	$5,98 \times 10^{24} \text{ kg}$	Charge on an electron	e	$-1,6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9,11 \times 10^{-31} \text{ kg}$	Avogadro's constant	N_A	$6,02 \times 10^{23} \text{ mol}^{-1}$

These are given to you in the exam. Everything in the formula sections below is given too — what is not given is which one to reach for, which is the whole exam.

→ MOTION IN ONE DIMENSION

Velocity	$v_f = v_i + a\Delta t$
Displacement	$\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2$
No time	$v_f^2 = v_i^2 + 2a\Delta x$
No acceleration	$\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t$
Vertical motion: replace Δx with Δy . Choose a positive direction first and keep it for the whole question — most lost marks here are sign errors, not method errors.	

⇄ FORCE, MOMENTUM & GRAVITATION

Newton's second law	$F_{net} = ma$
Weight	$w = mg$
Momentum	$p = mv$
Impulse	$F_{net} \Delta t = \Delta p = mv_f - mv_i$
Static friction	$f_s^{max} = \mu_s N$
Kinetic friction	$f_k = \mu_k N$
Newton's law of gravitation	$F = \frac{Gm_1m_2}{r^2}$
Gravitational field	$g = \frac{GM}{r^2}$
r is measured from the centre of the planet, so an object at height h has $r = R_E + h$.	

⚡ WORK, ENERGY & POWER

Work	$W = F\Delta x \cos \theta$
Gravitational PE	$U = mgh \quad (E_P = mgh)$
Kinetic energy	$K = \frac{1}{2}mv^2 \quad (E_K = \frac{1}{2}mv^2)$
Work–energy theorem	$W_{net} = \Delta K = K_f - K_i$
Non-conservative	$W_{nc} = \Delta K + \Delta U$
Power	$P = \frac{W}{\Delta t}$
Power at constant speed	$P_{av} = Fv_{av}$
θ is the angle between the force and the displacement. A force perpendicular to the motion does no work at all — that is why the normal force never appears in the energy sum.	

~ WAVES, SOUND & LIGHT

Wave equation	$v = f\lambda$
Period	$T = \frac{1}{f}$
Doppler effect	$f_L = \frac{v \pm v_L}{v \mp v_s} f_s$
Photon energy	$E = hf = \frac{hc}{\lambda}$
Photoelectric effect	$E = W_0 + E_{k(max)}$
Work function	$W_0 = hf_0$
Max. kinetic energy	$E_{k(max)} = \frac{1}{2}mv_{max}^2$
Doppler signs: choose the operator that makes the observed frequency rise when source and listener move towards each other. Substituting first and reasoning afterwards is how this question is lost.	

± ELECTROSTATICS

Coulomb's law	$F = \frac{kQ_1Q_2}{r^2}$
Field of a point charge	$E = \frac{kQ}{r^2}$
Field strength	$E = \frac{F}{q}$
Potential difference	$V = \frac{W}{q}$
Number of electrons	$n = \frac{Q}{e}$

Substitute charge magnitudes into Coulomb's law and get the direction from the diagram. Feeding signs into the formula gives the right number with the wrong direction about half the time.

~ ALTERNATING CURRENT

RMS current	$I_{rms} = \frac{I_{max}}{\sqrt{2}}$
RMS voltage	$V_{rms} = \frac{V_{max}}{\sqrt{2}}$
Average power	$P_{av} = V_{rms} I_{rms}$ $P_{av} = I_{rms}^2 R$ $P_{av} = \frac{V_{rms}^2}{R}$

A quoted mains voltage is always the rms value. The peak is $\sqrt{2}$ times larger.

⚗ PAPER 2 · CHEMISTRY FORMULAE

From mass	$n = \frac{m}{M}$
From particles	$n = \frac{N}{N_A}$
From gas volume (STP)	$n = \frac{V}{V_m}$
Concentration	$c = \frac{n}{V} = \frac{m}{MV}$
Titration	$\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$
pH	$\text{pH} = -\log[\text{H}_3\text{O}^+]$
Ionisation of water	$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1 \times 10^{-14}$
Cell potential	$E_{cell}^\theta = E_{cathode}^\theta - E_{anode}^\theta$
equivalently	$E_{cell}^\theta = E_{reduction}^\theta - E_{oxidation}^\theta$
or	$E_{cell}^\theta = E_{oxidising\ agent}^\theta - E_{reducing\ agent}^\theta$

c in $\text{mol}\cdot\text{dm}^{-3}$ means V must be in dm^3 . Millilitres are cubic centimetres; divide by 1000 before anything else. K_w is quoted at 298 K.

± ELECTRIC CIRCUITS

Ohm's law	$R = \frac{V}{I}$
Series	$R_s = R_1 + R_2 + \dots$
Parallel	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
Charge	$q = I\Delta t$
EMF and internal resistance	$\varepsilon = I(R + r)$

$$\text{Electrical energy} \quad W = Vq = VI\Delta t = I^2 R\Delta t = \frac{V^2 \Delta t}{R}$$

$$\text{Electrical power} \quad P = \frac{W}{\Delta t} = VI = I^2 R = \frac{V^2}{R}$$

Pick the power formula whose quantities you already have, rather than the one you remember first — all four are the same statement.

⚗ PAPER 2 · CHEMISTRY CONSTANTS

Quantity	Symbol	Value
Standard pressure	p^θ	$1,013 \times 10^5 \text{ Pa}$
Molar gas volume at STP	V_m	$22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$
Standard temperature	T^θ	273 K
Avogadro's constant	N_A	$6,02 \times 10^{23} \text{ mol}^{-1}$
Charge on an electron	e	$-1,6 \times 10^{-19} \text{ C}$

STP is 0°C and $101,3 \text{ kPa}$ — not room temperature. A question that says "at STP" is telling you to use $22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$.

☐ SI UNITS YOU ARE EXPECTED TO KNOW

Quantity	Unit	Quantity	Unit	Quantity	Unit
Force	newton, N	Energy / work	joule, J	Power	watt, W
Momentum	$\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$	Charge	coulomb, C	Potential difference	volt, V
Current	ampere, A	Resistance	ohm, Ω	Frequency	hertz, Hz
Electric field	$\text{N}\cdot\text{C}^{-1}$	Concentration	$\text{mol}\cdot\text{dm}^{-3}$	Gravitational field	$\text{N}\cdot\text{kg}^{-1}$

Units are not decoration: a final answer without one loses the mark it was carrying, and a unit that does not match the formula is the fastest way to catch your own arithmetic error before the marker does.

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